



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026-27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 4

Team No.: 14

Project Title: Linux Process Monitoring and Control System

Team Members

Roll Number	Student Name	Signature
2520030157	SAI SURENDRANATH PENDYALA	P. Suresh
2520030195	ADA ROHAN	Ada Rohan
2520030291	MEDIKONDA RAM RATAN	M. Ram Ratan

1. Abstract

Linux Process Monitoring and Control System is a Linux-based systems programming project designed to monitor and control running processes efficiently. The system provides information such as Process ID (PID), process state, CPU and memory usage, and other process-related details. It allows users to search, monitor, and control processes through a command-line interface. The project demonstrates important Operating Systems concepts including process management, process states, process identification, signals, process creation, and termination. Linux system interfaces and POSIX mechanisms are used to obtain process information and perform process control operations. The proposed system provides a simple and interactive environment for understanding how Linux manages processes and resources. The expected outcome is a functional process monitoring and control utility that helps users observe running processes and perform basic process management operations while demonstrating practical Operating Systems and Systems Programming concepts.

2. Problem Statement

Managing and monitoring multiple processes in a Linux environment can be difficult when users need to identify process details, observe process states, and control individual processes. Existing system utilities provide process information, but understanding their underlying Operating Systems mechanisms is important for Systems Programming. This project aims to develop a Linux Process Monitoring and Control System that allows users to view process information, monitor process activity, search for specific processes, and perform basic process control operations. The project focuses on demonstrating Linux process management, signals, system calls, and process-related interfaces.

3. Objectives

The project aims to monitor running Linux processes and display important information such as Process ID (PID), process state, CPU usage, and memory usage. It also provides process search and information features to help users identify and examine specific processes. The system implements process control operations such as sending signals and terminating selected processes. Additionally, the project demonstrates important Linux Operating Systems concepts, including process management, system calls, signals, and the /proc file system.

4. Proposed Methodology

The system will be developed as a command-line based Linux application using C. The application will obtain process-related information from Linux system interfaces, particularly the /proc file system and appropriate system calls. The user will interact with a menu-driven interface to perform operations such as listing processes, searching for a process, viewing process details, monitoring resource usage, and controlling processes. Process control operations will use Linux signals and appropriate system interfaces.

Basic Flow

Start → Display Menu → Select Operation → Obtain Process Information → Display/Control Process → Return to Menu → Exit

The methodology follows the Linux/POSIX - based approach required by your OSSP guidelines rather than making it a generic application.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
Process Management	Manage and monitor running processes
Process ID (PID)	Identify individual processes
Process States	Determine the current state of a process
Signals	Control or communicate with processes
kill()	Send signals to selected processes
fork()	Demonstrate process creation
wait() / waitpid()	Handle process termination/status
/proc filesystem	Obtain process and system information
System Calls	Interact with Linux OS services
POSIX APIs	Implement standard Linux process-management functionality

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520030157	SAI SURENDRA	Worked on ProcessManagement , PID and Process States.
2520030195	ADA ROHAN	Worked on Signals, kill() and fork()
2520030291	RAM RATAN	Worked on wait(), proc(), System Calls and POSIX APIs

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Ubuntu	Development and execution environment
C Programming	Core project implementation
GCC	Compilation of C programs
Linux / POSIX APIs	Process management and system programming
VS Code	Source code development
Git and GitHub	Version Control and project repository
Makefile	Build and Compilation management

8. Expected Outcome

The project is expected to provide a functional Linux Process Monitoring and Control System through a command-line interface. The system will allow users to view running processes, search for specific processes, display process information, monitor CPU and memory usage, and perform basic process control operations. The project will demonstrate practical understanding of Linux process management, signals, system calls, /proc filesystem, and POSIX mechanisms. It will also provide a working demonstration of how processes can be monitored and controlled in a Linux environment.

9. GitHub Repository

GitHub Repository Link: https://github.com/saisurendra0810/Team14_OSSP

Repository Created: Yes

Faculty Name and Signature

Faculty Name: E. Soumya

Signature: 

Remarks: _____

Faculty Use

Project Approved: YES

☒ Yes ☐ No

Date: 27/08/26

Faculty Signature: 